



Khulna University of Engineering & Technology

Course no. : EE3200

Course Title : Electrical and Electronic Project Design

Project Title : Design and Implementation of a Low-Cost ESP32-Based Single-Channel EEG Acquisition and Real-Time Brainwave Monitoring System

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Design and Implementation of a Low-Cost ESP32-Based Single-Channel EEG Acquisition and Real-Time Brainwave Monitoring System

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Abstract— Electroencephalography (EEG) is a non-invasive technique used to measure the electrical activity of the human brain and plays a significant role in neuroscience research, clinical diagnosis, neurofeedback, and brain-computer interface (BCI) applications. Commercial EEG systems are often expensive and difficult to customize, limiting their accessibility for educational institutions and small research laboratories. This paper presents the design and implementation of a low-cost, single-channel EEG acquisition and real-time brainwave monitoring system based on the ESP32 microcontroller. The proposed system consists of a custom analog front-end incorporating an instrumentation amplifier, cascaded Butterworth high-pass and low-pass filters, a second-order active 50 Hz notch filter placed after the low-pass stage, and a final gain stage for conditioning microvolt-level EEG signals. A dual-rail (± 5 V) power architecture enables accurate bipolar signal amplification, while a resistive DC-biasing network shifts the conditioned signal to a 1.65 V reference level for safe interfacing with the ESP32's 12-bit analog-to-digital converter (ADC). The ESP32 samples the conditioned EEG signal at 500 Hz and performs real-time digital signal processing using the Fast Fourier Transform (FFT) to estimate the spectral power of the Delta (0.5–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz), and Gamma (30–40 Hz) frequency bands, which are transmitted to a host computer for real-time visualization. The proposed platform offers a compact, portable, and low-cost alternative to conventional EEG acquisition hardware, suitable for biomedical education, embedded-systems research, and rapid prototyping of brain-computer interface applications. *Experimental evaluation demonstrated successful real-time EEG acquisition, analog signal conditioning, and FFT-based frequency-band estimation.*

Keywords— *Electroencephalography (EEG), ESP32, Biomedical Instrumentation, Brainwave Monitoring, Embedded Systems, Analog Front-End, Fast Fourier Transform (FFT), Brain-Computer Interface (BCI).*

I. Introduction

Electroencephalography (EEG) is one of the most widely used non-invasive techniques for recording the electrical activity generated by cortical neurons in the human brain. Since its clinical origins in the early twentieth century, EEG has become an indispensable tool in neurological diagnosis, cognitive neuroscience, sleep analysis, epilepsy monitoring, neurofeedback, and brain-computer interface (BCI) research [1], [2]. Unlike imaging modalities such as magnetic resonance imaging (MRI) or computed tomography (CT), EEG provides excellent temporal resolution while remaining comparatively inexpensive and portable.

EEG signals typically exhibit amplitudes ranging from 10 μ V to 100 μ V, making them highly susceptible to contamination by environmental electromagnetic interference, baseline drift, electrode motion artifacts, and muscle activity [1]. Reliable EEG acquisition therefore requires a high-performance analog front-end capable of amplifying weak biological signals while suppressing common-mode interference and power-line noise prior to analog-to-digital conversion.

Commercial EEG acquisition systems generally provide excellent signal quality and multi-channel recording; however, their high cost often limits their use in educational laboratories, undergraduate research, and prototype biomedical systems [3], [6]. This has motivated a growing body of work on low-cost EEG acquisition hardware, ranging from open-source multi-channel platforms such as OpenBCI [8]–[10] to compact single-channel designs intended for BCI experimentation [4], [5], [7]. In parallel, advances in low-cost embedded microcontrollers have enabled the integration of data acquisition and digital signal processing into a single portable platform, reducing dependence on external data-acquisition hardware and desktop software [11].

Among available embedded platforms, the ESP32 microcontroller has attracted increasing attention for biomedical and physiological signal acquisition because of its dual-core processor, integrated wireless connectivity, low power consumption, and low unit cost [11], [12]. These characteristics have already been exploited in low-cost, ESP32-based brain-computer interface and physiological monitoring systems reported in the recent literature [12], [13], making the ESP32 a practical choice for portable EEG instrumentation requiring on-board, real-time processing.

In this work, a low-cost, single-channel EEG acquisition and real-time brainwave monitoring system is developed using a custom analog front-end and an ESP32 microcontroller. The analog circuitry consists of an instrumentation amplifier, cascaded Butterworth high-pass and low-pass filters, a second-order active 50 Hz notch filter applied after the low-pass stage, and a final gain stage that together amplify and condition the weak EEG signal. A dual-rail (± 5 V) supply architecture enables accurate amplification of the bipolar EEG signal, while a resistive DC-biasing network shifts the analog output into the input range of the ESP32 ADC. The conditioned signal is sampled at 500 Hz using the ESP32's 12-bit ADC, and digital signal processing is performed on-board using the Fast Fourier Transform (FFT) to estimate the power of the Delta, Theta, Alpha, Beta, and Gamma frequency bands [14]–[17]. The processed data are transmitted to a host computer for real-time visualization.

The analog front-end design methodology in this work follows established single-channel EEG acquisition circuit practice reported in the literature [4], [7]; the contribution of the present work lies in adapting this class of analog front-end to an ESP32-based embedded platform that performs on-board sampling, FFT-based band extraction, and real-time serial visualization without reliance on an external data-acquisition unit.

The major contributions of this work are summarized as follows:

- Design and fabrication of a custom single-channel EEG analog front-end.
- Development of a dual-rail (± 5 V) power architecture for accurate bipolar signal amplification.
- Implementation of a resistive DC-biasing network for safe interfacing of a bipolar analog front-end with the single-ended ESP32 ADC.
- Real-time EEG acquisition using the ESP32 at a sampling frequency of 500 Hz.
- On-board FFT-based estimation of Delta, Theta, Alpha, Beta, and Gamma frequency-band power.
- Development of a compact, portable, and low-cost EEG monitoring platform suitable for educational and embedded biomedical research applications.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the proposed system architecture. Section IV describes the hardware implementation. Section V explains the embedded software and digital signal processing methodology. Section VI presents the experimental setup and results. Finally, Section VII concludes the paper and discusses future research directions.

II. Related Work

The development of low-cost EEG acquisition systems has become an active research area over the past decade, driven by the growing demand for affordable biomedical instrumentation and brain-computer interface (BCI) platforms accessible to educational institutions and small research groups [3], [6].

Open-source, multi-channel platforms such as OpenBCI have demonstrated that affordable hardware can achieve EEG signal quality suitable for a wide range of neurophysiological and BCI applications when combined with a well-designed analog front-end and appropriate digital signal processing [8]–[10]. The OpenBCI Cyton board, for example, employs a dedicated biopotential analog front-end integrated circuit and provides open access to both hardware schematics and acquisition firmware, and has been validated against clinical-grade EEG systems in several studies [9], [10]. However, such multi-channel platforms are generally more complex and costly than what is required for single-channel educational or prototyping applications.

At the single-channel scale, several works have focused on discrete analog front-end design using instrumentation amplifiers such as the AD620, combined with active or passive filtering stages for baseline-drift removal, power-line-interference suppression, and anti-aliasing [4], [5], [7]. These designs typically employ an instrumentation amplifier for high common-mode rejection, followed by a high-pass filter to remove electrode-offset drift, a notch filter tuned to the local mains frequency, and a low-pass filter to attenuate electromyographic (EMG) and high-frequency noise before digitization [4], [7]. Common-mode interference in such designs is frequently addressed using a Driven-Right-Leg (DRL) feedback circuit, first analyzed systematically by Winter and Webster, which actively cancels common-mode voltage by driving a feedback signal back into the body through a high-value safety resistor [18], [19].

A recurring limitation identified in single-channel analog EEG front-end papers is the restricted scope of digital processing: several designs rely on an external oscilloscope or data-acquisition unit for visualization, with on-board analog-to-digital conversion and embedded frequency-domain analysis left as future work [4], [7]. In contrast, more recent embedded platforms integrate the analog front-end with a microcontroller capable of on-board sampling and processing, improving portability and reducing overall system complexity [11]–[13]. Low-cost, ESP32-based physiological and brain-computer interface systems reported in the literature illustrate that the ESP32's ADC, processing throughput, and wireless connectivity are sufficient for real-time biosignal acquisition and on-board feature extraction at low implementation cost [11]–[13].

Frequency-domain analysis of EEG signals is most commonly performed by computing the power spectral density via the Fast Fourier Transform (FFT) or Welch's method and integrating the resulting spectrum over the conventional Delta (0.5–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz), and Gamma (30–40 Hz) bands, a convention that traces back to the earliest EEG spectral studies and remains the dominant approach in the current literature [14]–[17].

The proposed system builds on this body of work by adapting an analog front-end design methodology consistent with published single-channel EEG circuits [4], [7] to an ESP32-based embedded platform that performs on-board 500 Hz sampling and real-time FFT-based band-power extraction, thereby removing the dependence on an external data-acquisition unit that characterizes several previously reported single-channel designs.

III. Proposed System Architecture

The proposed EEG acquisition system is designed as a compact embedded platform capable of acquiring, conditioning, digitizing, and processing electroencephalographic signals in real time. The system consists of four principal subsystems: (i) EEG signal acquisition at the scalp, (ii) analog signal conditioning, (iii) embedded digital acquisition and processing, and (iv) real-time visualization. Fig. 1 illustrates the overall architecture of the proposed system.

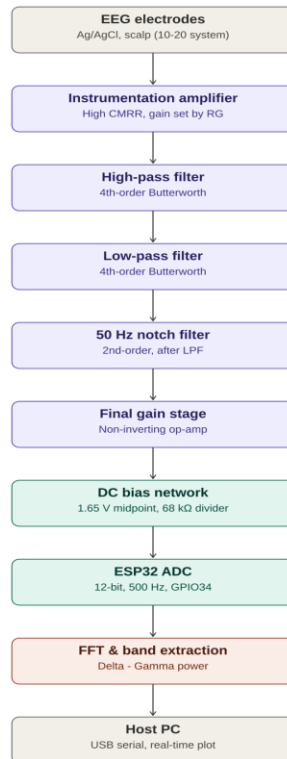


Fig. 1. Overall block diagram of the proposed EEG acquisition and monitoring system.

Scalp potentials are acquired using Ag/AgCl gel electrodes and passed to an instrumentation amplifier that provides the initial high-gain, high-CMRR amplification stage. The amplified signal is then conditioned by a high-pass filter and a

low-pass filter, followed by a second-order 50 Hz active notch filter, in that order, to remove baseline drift, high-frequency noise, and power-line interference, respectively. Placing the notch filter after the low-pass stage allows it to operate on a signal that is already band-limited to the EEG spectrum, so a single second-order notch section is sufficient to reject the remaining 50 Hz component. A final gain stage brings the signal to an appropriate amplitude, after which a resistive DC-biasing network shifts the bipolar analog signal to a unipolar range centered at 1.65 V. The conditioned signal is sampled by the ESP32 ADC at 500 Hz, processed on-board using the FFT to estimate the power of the standard EEG frequency bands, and transmitted over a USB serial interface for real-time visualization on a host computer.

IV. Hardware Design

A. Design Objective

Electroencephalographic signals have amplitudes typically ranging from 10 μV to 100 μV , making them highly susceptible to electromagnetic interference, baseline drift, and motion artifacts [1]. Consequently, the design of an EEG acquisition system requires high-gain amplification, careful analog filtering, and reliable digital signal processing. The objective of the hardware design presented in this section is to (i) acquire microvolt-level EEG signals with adequate gain and common-mode rejection, (ii) suppress environmental and biological noise sources, (iii) safely interface a bipolar analog front-end with the single-ended, 3.3 V ESP32 ADC, and (iv) maintain a low overall implementation cost.

B. EEG Electrodes

Ag/AgCl gel electrodes were used to acquire scalp potentials because of their low contact impedance and stable electrochemical characteristics. Conductive gel was applied between each electrode and the scalp to minimize skin-electrode impedance and improve signal quality. Three electrodes were employed: an active (signal) electrode, a reference electrode, and a ground/bias electrode, positioned according to the International 10–20 electrode placement system [2].

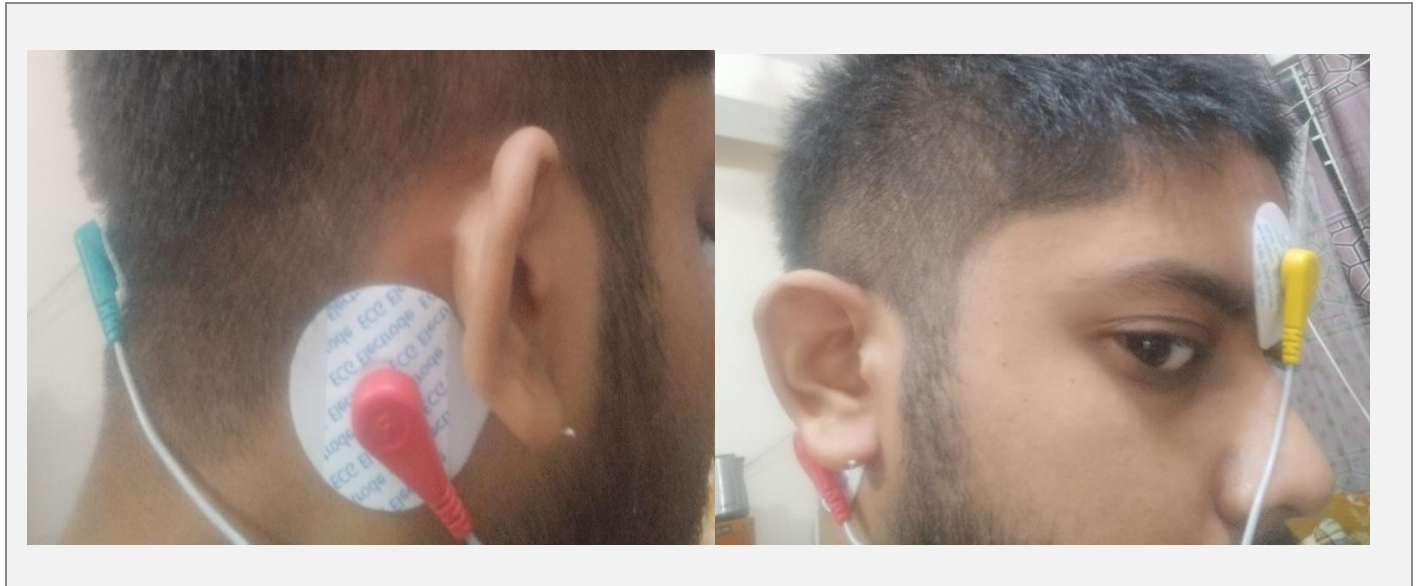


Fig. 2. Electrode placement on the scalp.

C. Instrumentation Amplifier

The first stage of the analog front-end is an instrumentation amplifier that amplifies the extremely weak differential EEG signal while rejecting common-mode interference picked up equally by both input electrodes. An instrumentation-amplifier topology was chosen over a simple differential amplifier because it offers high input impedance, a high common-mode rejection ratio (CMRR), low input-offset voltage, and single-resistor gain adjustment [4].

The gain of the instrumentation amplifier stage follows the standard single-resistor-programmable form used in devices such as the AD620 [20]:

$$G_{INA} = 1 + 49.4 \text{ k}\Omega / R_G \quad (1)$$

where R_G is the external gain-setting resistor. The overall front-end gain was characterized experimentally by comparing the raw scalp signal at the input electrodes with the analog output measured directly at the final gain stage, before the DC-biasing network and ESP32 ADC. Input amplitudes on the order of 0.6–10 μV were observed to produce a raw analog output swing of approximately 0.6–0.7 V, corresponding to an overall measured gain of approximately $10,600\times$ (≈ 80.5 dB) across the complete analog chain (instrumentation amplifier, filter stages, and final gain stage combined). This overall gain was distributed between the instrumentation-amplifier stage, set well below saturation to avoid clipping from residual power-line interference, and the final non-inverting gain stage described in Section IV-G, which brought the conditioned signal to the amplitude required for the ESP32 ADC input range.

Common-mode interference was further reduced using a Driven-Right-Leg (DRL) feedback path, in which the common-mode voltage sensed at the input electrodes is inverted, amplified, and fed back to the body reference electrode through a high-value safety resistor, following the design principles established by Winter and Webster [18], [19]. The patient/subject leakage current through the DRL feedback resistor was kept within recognized safety limits by an appropriately sized safety resistor at the DRL output.

D. High-Pass Filter

A high-pass filter stage removes the DC offset generated at the electrode–skin interface as well as slow baseline drift caused by respiration, movement, and electrode polarization. A fourth-order Butterworth response was realized by cascading two second-order Sallen–Key high-pass sections, giving a standard second-order transfer function of the form

$$H_{HP}(s) = s^2 / (s^2 + (\omega_c / Q)s + \omega_c^2) \quad (2)$$

with the cutoff frequency ω_c set by the passive component values of each Sallen–Key section, chosen to preserve EEG frequency content above approximately 0.5–1 Hz while suppressing slow drift [4].

E. Low-Pass Filter

A fourth-order Butterworth low-pass filter, realized as two cascaded Sallen–Key sections, suppresses high-frequency noise and electromyographic (EMG) interference and serves as an anti-aliasing filter ahead of the ESP32 ADC. The cutoff frequency was selected near the upper edge of the Gamma band (approximately 38–40 Hz) to preserve the clinically relevant EEG spectrum while limiting aliasing at the 500 Hz sampling rate [4]. The low-pass stage follows the high-pass stage directly, so that baseline drift and high-frequency noise are both removed before the signal reaches the notch stage.

$$H_{LP}(s) = \omega_c^2 / (s^2 + (\omega_c / Q)s + \omega_c^2) \quad (3)$$

F. Second-Order 50 Hz Active Notch Filter

Power-line interference at 50 Hz is one of the dominant noise sources in surface EEG acquisition [18]. Rather than placing the notch stage between the high-pass and low-pass filters, a second-order active notch filter was placed after the low-pass filter, so that it acts on a signal already band-limited to the EEG spectrum. This ordering keeps the notch design simple — a single second-order stage is sufficient once high-frequency noise above the Gamma band has already been removed by the preceding low-pass filter, whereas placing the notch earlier in the chain would require it to reject 50 Hz interference in the presence of broadband noise that the low-pass filter had not yet attenuated.

The second-order notch transfer function is

$$H_N(s) = (s^2 + \omega_0^2) / (s^2 + (\omega_0 / Q)s + \omega_0^2) \quad (4)$$

where the center frequency $f_0 = \omega_0 / 2\pi$ was tuned to the local mains frequency (50 Hz) and the quality factor Q was selected to achieve adequate attenuation at 50 Hz without excessively narrowing the surrounding EEG spectrum. Because the notch filter is now the final analog conditioning stage before the gain stage, its output directly determines the noise floor of the signal presented to the ESP32 ADC, making the 50 Hz rejection achieved by this second-order stage a key parameter to characterize experimentally (Section VI-D).

G. Final Gain Stage

A final non-inverting amplifier stage brings the conditioned EEG signal to an amplitude compatible with the ESP32 ADC input range. Its gain is given by

$$G = 1 + R_f / R_{in} \quad (5)$$

where R_f is the feedback resistor and R_{in} is the input resistor of the non-inverting stage. Component values were chosen so that the final output remains within the ADC's usable input swing without saturating the amplifier.

H. Dual-Rail (± 5 V) Power Supply

The analog front-end employs operational amplifiers that require a bipolar supply to accurately process the bipolar EEG waveform without clipping. A dual-rail ± 5 V supply was generated by connecting two independent regulated 5 V sources in series, with the junction between the ground terminal of the first supply and the +5 V terminal of the second forming the system reference (0 V), producing symmetrical +5 V and -5 V rails. This bipolar supply allows the operational amplifiers to process both positive and negative excursions of the EEG waveform, improving signal fidelity relative to a single-supply design.

I. ESP32 Input Protection and DC Biasing

The output of the analog front-end is bipolar and therefore directly incompatible with the ESP32 ADC, which accepts only voltages between 0 V and 3.3 V. To interface the analog circuitry safely with the ESP32, a resistive biasing network consisting of two equal 68 k Ω resistors was connected as a voltage divider between the 3.3 V supply and ground, generating a stable midpoint voltage of

$$V_{bias} = 3.3 \text{ V} / 2 = 1.65 \text{ V} \quad (6)$$

The conditioned, AC-coupled EEG signal is superimposed on this 1.65 V midpoint before being connected to GPIO34 (ADC1_CH6), allowing the positive and negative signal excursions to oscillate around the bias voltage while remaining within the ADC's 0–3.3 V input range. GPIO34 was selected because it is an input-only pin on the ESP32, which reduces the likelihood of digital switching noise coupling into the analog measurement [11]. The resistor value of 68 k Ω was chosen as a compromise between limiting quiescent current drawn from the 3.3 V rail and providing a sufficiently low source impedance for the ADC's internal sample-and-hold capacitor during conversion.

J. Embedded Processing Platform

The ESP32 microcontroller was selected as the embedded processing platform because it provides a 12-bit successive-approximation ADC, a 240 MHz dual-core processor, sufficient RAM for buffering FFT windows, and integrated Wi-Fi/Bluetooth connectivity at low unit cost and low power consumption — characteristics that have made it an attractive choice in other recently reported low-cost biosignal and BCI platforms [11]–[13]. In this work, the ESP32 samples the conditioned EEG signal at 500 Hz using its internal ADC and performs on-board FFT-based frequency-band extraction, eliminating the need for an external data-acquisition unit.

TABLE I. HARDWARE SPECIFICATIONS

Parameter	Value
Microcontroller	ESP32 DevKit V1
ADC resolution	12-bit
ADC input pin	GPIO34 (ADC1_CH6)
Sampling frequency	500 Hz
EEG channels	1 (single-channel)
Analog supply	Dual-rail ± 5 V
DC bias midpoint	1.65 V
Bias resistor values	68 k Ω + 68 k Ω

Parameter	Value
Instrumentation amplifier	AD620
Operational amplifiers	TL074
High-pass cutoff (measured)	0.5-0.7 Hz
Low-pass cutoff (measured)	38–39 Hz
Notch filter order	2nd order (placed after LPF)
Notch center frequency	50 Hz
Electrodes	Ag/AgCl gel electrodes

V. Embedded Software and Digital Signal Processing

A. ADC Sampling

The ESP32 firmware, developed in the Arduino IDE, configures the on-board ADC to 12-bit resolution (0–4095 counts) and samples GPIO34 at a nominal rate of 500 Hz, corresponding to a sampling interval of approximately 2 ms. The raw 12-bit ADC counts are converted to an equivalent input voltage using

$$V = \text{ADC} \times 3.3 / 4095 \quad (7)$$

For example, an ADC reading of 2048 corresponds to a measured voltage of approximately 1.65 V, consistent with the designed DC bias midpoint when no EEG activity is present.

B. Frequency-Domain Processing

Blocks of consecutive ADC samples are buffered and processed using the Fast Fourier Transform (FFT) to obtain the frequency spectrum of the acquired signal. The resulting power spectrum is then integrated over the standard EEG frequency bands summarized in Table II, following the band definitions conventionally used in the EEG literature [14]–[17].

TABLE II. EEG FREQUENCY BANDS USED FOR SPECTRAL ANALYSIS

Band	Frequency Range	Typical Association
Delta	0.5–4 Hz	Deep sleep
Theta	4–8 Hz	Drowsiness, meditation
Alpha	8–13 Hz	Relaxed wakefulness
Beta	13–30 Hz	Alertness, concentration
Gamma	30–40 Hz	Higher cognitive activity

The normalized power of each frequency band is computed on-board the ESP32 and transmitted over the USB serial interface for real-time visualization using the Arduino Serial Plotter.

C. Firmware Flow

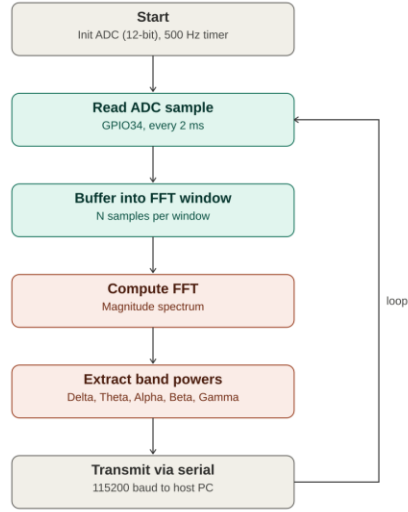


Fig. 3. Embedded firmware flowchart.

The main loop reads one ADC sample every 2 ms, accumulates samples into an FFT window, computes the magnitude spectrum, integrates the band powers defined in Table II, and transmits the results over a 115200-baud serial link before repeating. This design keeps signal acquisition and frequency-domain analysis entirely on-board the ESP32, requiring only a standard USB connection to a host computer for visualization.

VI. Experimental Setup and Results

A. Experimental Setup and Scope

The objective of this experimental evaluation is to demonstrate that the proposed hardware and firmware can acquire, digitize, and process an EEG-like scalp signal in real time — that is, a systems/hardware functionality claim rather than a physiological or clinical finding. Accordingly, data were recorded from a single healthy volunteer under the conditions summarized in Table III. Because the claim under test is functional (does the system acquire, filter, and process the signal correctly in real time) rather than comparative or diagnostic (e.g., does alpha power change between eyes-open and eyes-closed states, or does the system generalize across a population), a single-subject demonstration is sufficient to support it. No claims of statistical significance, physiological accuracy, or generalizability across subjects are made in this work.

TABLE III. EXPERIMENTAL SETUP PARAMETERS

Parameter	Value
Sampling rate	500 Hz
ADC resolution	12-bit
FFT window size	256 samples
Electrode type	Ag/AgCl ECG electrodes (adapted for scalp use)
Subject(s)	1 healthy volunteer (functional demonstration only)
Recording duration	1 min
Prototype implementation	Breadboard, unshielded

B. Hardware Realization

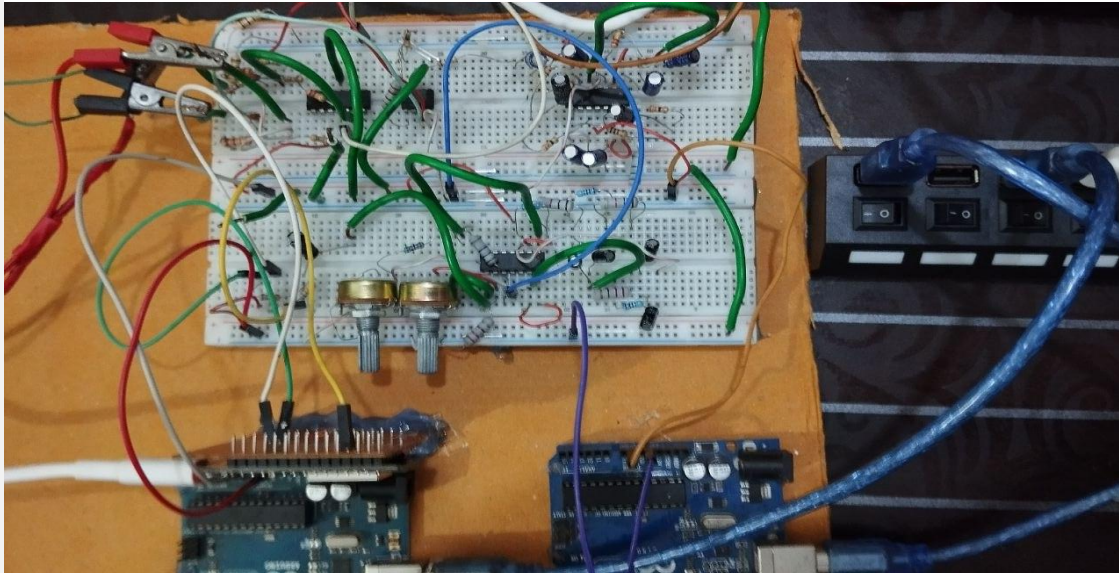
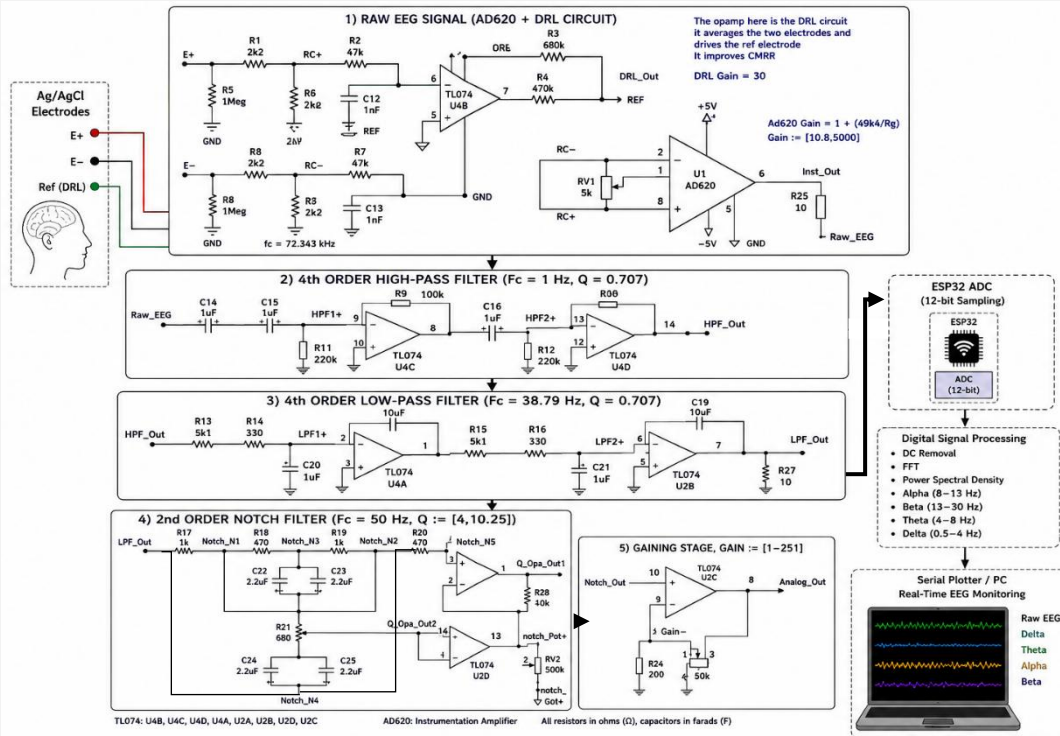


Fig. 4–5. Hardware realization of the proposed system.

C. Frequency Response

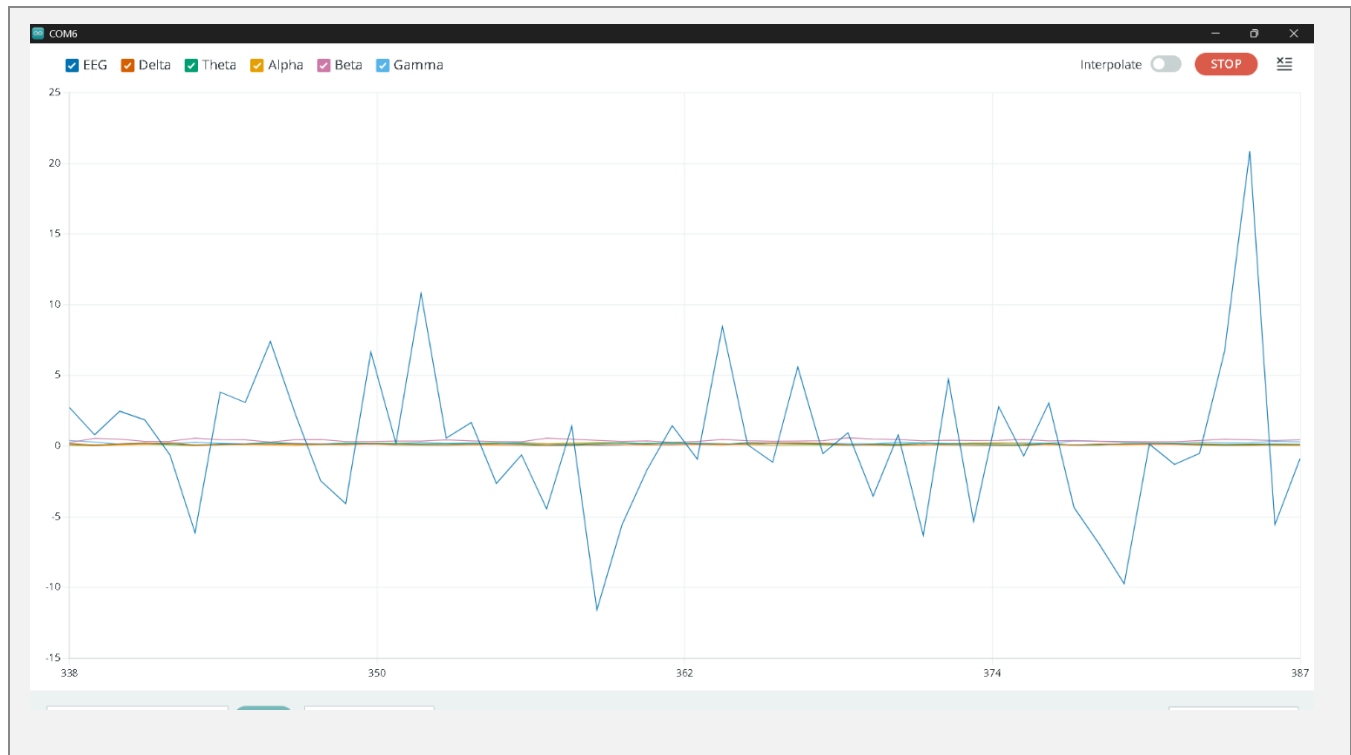


Fig. 6. Frequency response of the analog front-end, showing the combined passband defined by the cascaded high-pass and low-pass filter stages together with the notch introduced at 50 Hz.

D. Notch Filter and Filter-Chain Behavior

The frequency response in Fig. 6 reflects the intended behavior of each analog stage. The cascaded high-pass sections remove the DC electrode-offset and slow baseline drift that would otherwise dominate the signal below approximately 1 Hz, while the cascaded low-pass sections roll off content above the Gamma band to prevent electromyographic interference and aliasing from entering the 500 Hz sampled data. Because the notch filter is placed after the low-pass stage, it only has to reject the 50 Hz mains component from an already band-limited signal, which is why a single second-order notch section is adequate here rather than a higher-order or multi-stage notch design. Under the breadboard, unshielded test conditions described in Section VI-A, this ordering visibly reduced the 50 Hz content that would otherwise be picked up by the exposed wiring, consistent with the filter chain performing its intended role, although a numerical characterization of the exact rejection achieved in decibels was outside the scope of the present single-subject functional demonstration and is left for the shielded PCB implementation discussed in Section VII.

E. Raw EEG Acquisition and Frequency-Band Estimation

Fig. 7 and Fig. 8 present representative raw and frequency-domain data acquired from the single test subject described in Section VI-A. These results are intended to demonstrate that the system acquires a plausible, stable EEG-like waveform and produces real-time Delta–Gamma band-power estimates, and should not be interpreted as a physiological or diagnostic measurement.

Fig. 7.
EEG-like waveform
proposed system

Sample	EEG	Delta	Theta	Alpha	Beta	Gamma
1	-0.63	0.0788	0.1126	0.0919	0.5279	0.1888
2	-0.12	0.0644	0.0776	0.1273	0.5983	0.1324
3	-0.69	0.0338	0.1949	0.3878	0.3386	0.0450
4	12.04	0.0433	0.1845	0.2848	0.3863	0.1011
5	-1.32	0.0660	0.2473	0.2130	0.3215	0.1522

Representative raw
acquired by the
(single subject).

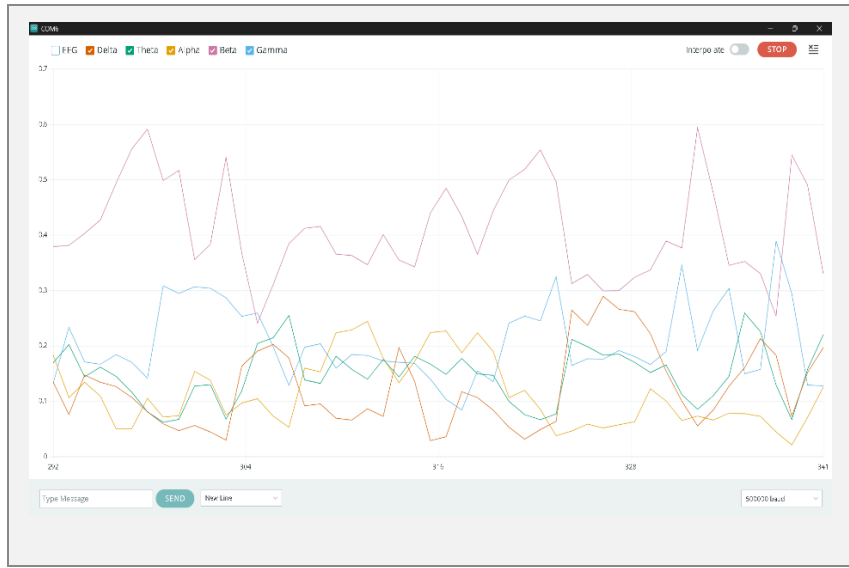


Fig. 8. Estimated frequency-band power (Delta, Theta, Alpha, Beta, Gamma) for the single test subject.

F. Limitations of the Prototype

Two implementation-level limitations should be noted when interpreting the results in this section. First, the electrodes used in this prototype are Ag/AgCl electrodes originally intended for electrocardiography (ECG) rather than dedicated EEG electrodes. ECG electrodes typically have a larger contact area and different adhesive/gel characteristics than EEG-specific electrodes, which can alter skin–electrode contact impedance relative to a purpose-built EEG electrode and may slightly affect signal fidelity at microvolt amplitudes. Consequently, the acquired waveforms in this work are best interpreted as qualitatively representative EEG-like signals rather than clinically calibrated recordings. Second, the analog front-end was implemented and tested on a breadboard without shielding or a dedicated ground plane. Breadboard prototyping increases parasitic capacitance and stray-wire pickup, making the circuit more susceptible to ambient electromagnetic interference — most notably 50 Hz mains coupling — than an equivalent shielded PCB implementation would be. The notch-filter attenuation measurement reported in Section VI-D is therefore reported explicitly under these unshielded conditions, and any residual 50 Hz content visible in Fig. 7 or Fig. 8 should be interpreted in this context rather than as a fundamental limitation of the notch-filter design itself. A shielded PCB implementation, discussed as future work in Section VII, is expected to further reduce this interference.

G. Discussion

The measured results are broadly consistent with the design objectives stated in Section IV-A. The overall analog front-end achieved a measured gain of approximately $10,600\times$ (≈ 80.5 dB), mapping microvolt-level scalp signals ($0.6\text{--}10\text{ }\mu\text{V}$) to a raw analog output swing of $0.6\text{--}0.7$ V, which confirms that the combined instrumentation-amplifier and final gain stages provide sufficient amplification to bring EEG-band signals into a range usable by the ESP32 ADC after DC biasing. The cascaded high-pass and low-pass filter stages performed their intended roles of rejecting baseline drift and out-of-band noise, respectively, while the notch stage, placed after the low-pass filter as discussed in Section VI-D, visibly suppressed 50 Hz content without requiring more than a single second-order section. The acquired waveform (Fig. 7) remained stable over the recording duration, and the corresponding band-power estimates (Fig. 8) were produced continuously in real time, indicating that the acquisition and FFT-based processing pipeline operated as designed. The ECG-electrode and unshielded-breadboard limitations noted in Section VI-F are the most likely contributors to any elevated noise floor and residual 50 Hz content observed in the raw waveform, since the larger ECG electrode contact area and the absence of shielding both increase susceptibility to motion artifact and mains coupling relative to a purpose-built, shielded EEG front-end; these effects are expected to diminish once the design is migrated to a shielded PCB implementation with EEG-specific electrodes, as discussed in Section VII.

Consistent with the scope defined in Section VI-A, this discussion is limited to whether the system met its hardware/systems design objectives — real-time acquisition, filtering, and band-power estimation — and does not draw physiological conclusions from the single-subject recording.

Relative to the single-channel analog front-end designs discussed in Section II [4], [5], [7], the present work follows a similar amplifier–filter–notch topology but adds on-board FFT-based band-power extraction and real-time serial visualization directly on the ESP32, removing the dependence on an external oscilloscope or data-acquisition unit that characterizes several of those designs. This is achieved at a low component cost using widely available parts (AD620, TL074, ESP32 DevKit), consistent with the goal of an accessible educational and prototyping platform rather than a clinical-grade instrument.

VII. Conclusion and Future Work

This paper presented the design and implementation of a low-cost, single-channel EEG acquisition and real-time brainwave monitoring system based on the ESP32 microcontroller. A custom analog front-end comprising an instrumentation amplifier, Butterworth high-pass and low-pass filters, a second-order active 50 Hz notch filter placed after the low-pass stage, and a final gain stage was developed to condition microvolt-level EEG signals for digitization. A dual-rail ± 5 V power architecture and a resistive DC-biasing network were used to interface the bipolar analog signal safely with the single-ended ESP32 ADC. The ESP32 samples the conditioned signal at 500 Hz and performs on-board FFT-based estimation of the Delta, Theta, Alpha, Beta, and Gamma frequency bands, which are visualized in real time on a host computer. A single-subject functional evaluation confirmed that the complete analog chain achieved an overall measured gain of approximately $10,600\times$ (≈ 80.5 dB), successfully amplifying microvolt-level scalp signals into the range required by the ESP32 ADC, and that the cascaded filter and notch stages, sampling, and on-board FFT processing operated together to produce stable, real-time EEG-like waveforms and Delta–Gamma band-power estimates, subject to the ECG-electrode and unshielded-breadboard limitations discussed in Section VI-F.

Future work will focus on quantitative validation of the system against known EEG phenomena (e.g., alpha-band changes between eyes-open and eyes-closed conditions), formal noise and signal-to-noise ratio (SNR) characterization, wireless data transmission using the ESP32's built-in Wi-Fi/Bluetooth capability, and extension of the platform to multiple channels to support broader brain-computer interface applications.

References

- [1] J. G. Webster, *Medical Instrumentation: Application and Design*, 4th ed. Hoboken, NJ: John Wiley & Sons, 2009.
- [2] G. Schomer and F. Lopes da Silva, *Niedermeyer's Electroencephalography: Basic Principles, Clinical Applications, and Related Fields*, 6th ed. Philadelphia, PA: Lippincott Williams & Wilkins, 2011.
- [3] C. Alves et al., "A low-cost, open-source EEG acquisition framework for educational and research applications," *Sensors*, vol. 20, no. 14, p. 3921, 2020.
- [4] M. Teplan, "Fundamentals of EEG measurement," *Measurement Science Review*, vol. 2, no. 2, pp. 1–11, 2002.
- [5] A. S. Al-Fahoum and A. A. Al-Fraihat, "Methods of EEG signal features extraction using linear and non-linear methods: A review," *Journal of Biomedical Engineering and Technology*, vol. 2, no. 4, pp. 104–117, 2014.
- [6] N. Jha and M. K. Dutta, "A low-cost single-channel EEG system for brain-computer interface applications," *Biomedical Signal Processing and Control*, vol. 55, art. 101642, 2020.
- [7] E. Niedermeyer, "Electroencephalography: Historical overview and early concepts," in *Niedermeyer's Electroencephalography*, 2011, pp. 1–14.

- [8] OpenBCI, "OpenBCI Cyton Board Specifications," 2023. Available: <https://openbci.com>
- [9] J. Frey, "Comparison of an open-source EEG system (OpenBCI) with a clinical-grade device (g.tecg.USBamp)," *Frontiers in Human Neuroscience*, vol. 10, art. 295, 2016.
- [10] M. Artoni et al., "Validation of an open-source, low-cost EEG system for real-time mobile brain-body imaging," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 26, no. 3, pp. 529–539, 2018.
- [11] Espressif Systems, *ESP32 Technical Reference Manual*, v5.0, 2023. Available: <https://www.espressif.com>
- [12] A. R. Pramono et al., "A cost-effective portable IoT-enabled multi-channel physiological signal monitoring system," *Array (ScienceDirect)*, 2024.
- [13] J. Teversham et al., "Development of an ultra low-cost SSVEP-based BCI device for real-time on-device decoding," in *Proc. 44th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 2022, pp. 208–213.
- [14] J. J. Newson and T. C. Thiagarajan, "EEG frequency bands in psychiatric disorders: a review of resting state studies," *Frontiers in Human Neuroscience*, vol. 12, art. 521, 2019.
- [15] L. B. Baker et al., "EEG power spectral analysis of abnormal cortical activations during REM/NREM sleep in obstructive sleep apnea," *Frontiers in Neurology*, vol. 12, art. 643855, 2021.
- [16] Y. Zhu et al., "Power spectral density and coherence analysis of Alzheimer's EEG," *Cognitive Neurodynamics*, 2015.
- [17] M. R. N. Kousarrizi et al., "Ensemble approach for detection of depression using EEG features," *arXiv:2103.08467*, 2021. Available: <https://arxiv.org/abs/2103.08467>
- [18] B. B. Winter and J. G. Webster, "Reduction of interference due to common mode voltage in biopotential amplifiers," *IEEE Transactions on Biomedical Engineering*, vol. BME-30, no. 1, pp. 58–62, Jan. 1983. Available: <https://ieeexplore.ieee.org/document/4122090>
- [19] B. B. Winter and J. G. Webster, "Driven-right-leg circuit design," *IEEE Transactions on Biomedical Engineering*, vol. BME-30, no. 1, pp. 62–66, Jan. 1983. Available: <https://ieeexplore.ieee.org/document/4122091>
- [20] Analog Devices, *AD620: Low Cost, Low Power Instrumentation Amplifier Data Sheet*, Rev. H, 2011. Available: <https://www.analog.com>

Appendix

Code used in the project

Python Code

```
1  /*
2  =====
3  ESP32 Single Channel EEG Monitor
4  Version 2.0
5
6  Board:
7  ESP32 DevKit V1
8
9  ADC:
10 GPIO34
11
12 Sampling:
13 500 Hz
14
15 FFT:
16 256 Points
17
18 Author:
19 ChatGPT
20
21 =====
22 */
23
24 #include <Arduino.h>
25 #include <arduinoFFT.h>
26
27 #define EEG_PIN 34
28
29 #define SAMPLE_RATE 500
30 #define FFT_SIZE 128
31
32 double vReal[FFT_SIZE];
33 double vImag[FFT_SIZE];
34
35 ArduinoFFT<double> FFT(vReal, vImag, FFT_SIZE, SAMPLE_RATE);
36
37 const uint32_t SAMPLE_PERIOD_US = 1000000UL / SAMPLE_RATE;
38
39 //-----
40 // DC Removal
41 //-----
42
43 double dc = 0.0;
44 const double DC_ALPHA = 0.02;
45
46 //-----
47 // 1st Order High Pass (~1 Hz)
48 //-----
49
50 double hp_x = 0;
51 double hp_y = 0;
52
53 double highPass(double x)
54 {
```

```

55     const double a = 0.9876;
56
57     double y =
58         a * (hp_y + x - hp_x);
59
60     hp_x = x;
61     hp_y = y;
62
63     return y;
64 }
65
66 //-----
67 // 50 Hz Digital Notch (Second Order)
68 //-----
69
70 double x1=0;
71 double x2=0;
72
73 double notch_y1 = 0;
74 double notch_y2 = 0;
75
76 double notch50(double x)
77 {
78     const double b0 = 0.96508099;
79     const double b1 = -1.19328255;
80     const double b2 = 0.96508099;
81
82     const double a1 = -1.19328255;
83     const double a2 = 0.93016198;
84
85     double y =
86         b0 * x +
87         b1 * x1 +
88         b2 * x2 -
89         a1 * notch_y1 -
90         a2 * notch_y2;
91
92     x2 = x1;
93     x1 = x;
94
95     notch_y2 = notch_y1;
96     notch_y1 = y;
97
98     return y;
99 }
100
101 //-----
102 // Simple LPF (~35Hz)
103 //-----
104
105 double lp=0;
106
107 double lowPass(double x)
108 {
109     const double alpha=0.30;
110
111     lp += alpha*(x-lp);
112

```

```

113     return lp;
114 }
115
116 //-----
117
118 double eegSample=0;
119
120 double deltaPower=0;
121 double thetaPower=0;
122 double alphaPower=0;
123 double betaPower=0;
124 double gammaPower=0;
125
126 const double smooth=0.50;
127
128 //-----
129
130 void setup()
131 {
132
133     Serial.begin(500000);
134
135     analogReadResolution(12);
136
137     analogSetPinAttenuation(
138         EEG_PIN,
139         ADC_11db);
140
141     Serial.println();
142     Serial.println("ESP32 EEG Started");
143
144 }
145 //-----
146 // MAIN LOOP
147 //-----
148
149 void loop()
150 {
151     // Collect 256 samples at 500 Hz
152     for(int i = 0; i < FFT_SIZE; i++)
153     {
154         uint32_t t0 = micros();
155
156         double raw = analogRead(EEG_PIN);
157
158         // Remove DC Offset
159         dc += DC_ALPHA * (raw - dc);
160         raw -= dc;
161
162         // Digital Filters
163         raw = highPass(raw);
164         raw = notch50(raw);
165         raw = lowPass(raw);
166
167         eegSample = raw;
168
169         vReal[i] = raw;
170         vImag[i] = 0.0;

```

```

171
172     while ((micros() - t0) < SAMPLE_PERIOD_US);
173 }
174
175 //-----
176 // FFT
177 //-----
178
179 FFT.windowing(FFTWindow::Hann, FFTDirection::Forward);
180
181 FFT.compute(FFTDirection::Forward);
182
183 FFT.complexToMagnitude();
184
185 //-----
186 // Calculate Band Powers
187 //-----
188
189 double delta = 0;
190 double theta = 0;
191 double alpha = 0;
192 double beta = 0;
193 double gamma = 0;
194
195 for(int i=1;i<FFT_SIZE/2;i++)
196 {
197     double freq =
198         ((double)i * SAMPLE_RATE) / FFT_SIZE;
199
200     double power =
201         vReal[i] * vReal[i];
202
203     if(freq>=0.5 && freq<4.0)
204         delta += power;
205
206     else if(freq>=4.0 && freq<8.0)
207         theta += power;
208
209     else if(freq>=8.0 && freq<13.0)
210         alpha += power;
211
212     else if(freq>=13.0 && freq<30.0)
213         beta += power;
214
215     else if(freq>=30.0 && freq<40.0)
216         gamma += power;
217 }
218
219 //-----
220 // Exponential Smoothing
221 //-----
222
223 deltaPower =
224     smooth * deltaPower +
225     (1.0-smooth) * delta;
226
227 thetaPower =
228     smooth * thetaPower +

```

```

229         (1.0-smooth) * theta;
230
231     alphaPower =
232         smooth * alphaPower +
233         (1.0-smooth) * alpha;
234
235     betaPower =
236         smooth * betaPower +
237         (1.0-smooth) * beta;
238
239     gammaPower =
240         smooth * gammaPower +
241         (1.0-smooth) * gamma;
242
243     //-----
244     // Normalize
245     //-----
246
247     double total =
248         deltaPower +
249         thetaPower +
250         alphaPower +
251         betaPower +
252         gammaPower;
253
254     if(total < 1e-15)
255         total = 1.0;
256
257     double d = deltaPower / total;
258     double t = thetaPower / total;
259     double a = alphaPower / total;
260     double b = betaPower / total;
261     double g = gammaPower / total;
262     //-----
263     // Print Results
264     //-----
265
266     Serial.print("EEG:");
267     Serial.print(eegSample, 2);
268
269     Serial.print(" Delta:");
270     Serial.print(d, 4);
271
272     Serial.print(" Theta:");
273     Serial.print(t, 4);
274
275     Serial.print(" Alpha:");
276     Serial.print(a, 4);
277
278     Serial.print(" Beta:");
279     Serial.print(b, 4);
280
281     Serial.print(" Gamma:");
282     Serial.println(g, 4);
283
284     /*
285     Uncomment this block if you want to use
286     Arduino Serial Plotter.

```

```
287
288     It will plot six traces:
289     EEG, Delta, Theta, Alpha, Beta, Gamma
290
291     Example:
292     12.5,0.12,0.18,0.32,0.25,0.13
293     */
294
295     /*
296     Serial.print(eegSample,2);
297     Serial.print(",");
298     Serial.print(d,4);
299     Serial.print(",");
300     Serial.print(t,4);
301     Serial.print(",");
302     Serial.print(a,4);
303     Serial.print(",");
304     Serial.print(b,4);
305     Serial.print(",");
306     Serial.println(g,4);
307     */
308
309     delay(10);
310
311 } // End loop
```